

**2** Two quantities are measured.

$$L = 6.8 \pm 0.1 \text{ cm}$$

$$T = 2.42 \pm 0.08 \text{ s}$$

$L$  and  $T$  are related to  $X$  by the equation shown.

$$X = \frac{4\pi^2 L}{T^2}$$

What is the calculated value and uncertainty of  $X$ ?

**A**  $45.8 \pm 0.2 \text{ cm s}^{-2}$

**B**  $45.8 \pm 0.3 \text{ cm s}^{-2}$

**C**  $46 \pm 2 \text{ cm s}^{-2}$

**D**  $46 \pm 4 \text{ cm s}^{-2}$